

Bloch sphere

Let $|s\rangle$ be the spin state

$$|s\rangle = \begin{pmatrix} \cos(\theta/2) \\ \sin(\theta/2) \exp(i\phi) \end{pmatrix}$$

The expectation values for $|s\rangle$ are

$$\langle S_x \rangle = \langle s | S_x | s \rangle = \frac{\hbar}{2} \sin \theta \cos \phi$$

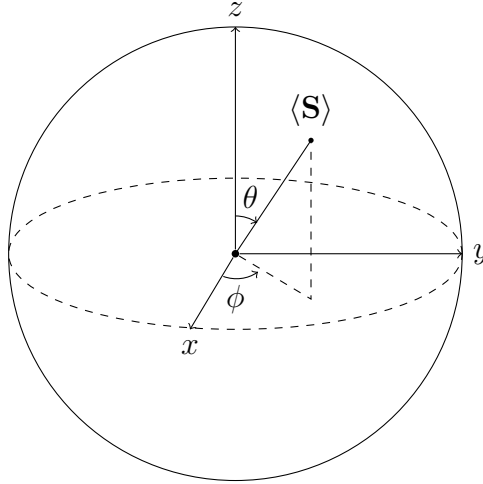
$$\langle S_y \rangle = \langle s | S_y | s \rangle = \frac{\hbar}{2} \sin \theta \sin \phi$$

$$\langle S_z \rangle = \langle s | S_z | s \rangle = \frac{\hbar}{2} \cos \theta$$

In vector form

$$\langle \mathbf{S} \rangle = \langle s | \mathbf{S} | s \rangle = \frac{\hbar}{2} \begin{pmatrix} \sin \theta \cos \phi \\ \sin \theta \sin \phi \\ \cos \theta \end{pmatrix}$$

Hence $\langle \mathbf{S} \rangle$ is a point on a sphere of radius $\hbar/2$.



To show completeness, start with an arbitrary complex vector $|\chi\rangle$ such that $\langle \chi | \chi \rangle = 1$.

$$|\chi\rangle = \begin{pmatrix} A \exp(i\phi_1) \\ \sqrt{1 - A^2} \exp(i\phi_2) \end{pmatrix}$$

Substitute $\cos(\theta/2)$ for A and $\sin(\theta/2)$ for $\sqrt{1 - A^2}$.

$$|\chi\rangle = \begin{pmatrix} \cos(\theta/2) \exp(i\phi_1) \\ \sin(\theta/2) \exp(i\phi_2) \end{pmatrix}$$

Let $|s\rangle$ be the following phase of $|\chi\rangle$.

$$|s\rangle = \exp(-i\phi_1) |\chi\rangle = \begin{pmatrix} \cos(\theta/2) \\ \sin(\theta/2) \exp[i(\phi_2 - \phi_1)] \end{pmatrix}$$

By equality of expectation every $|\chi\rangle$ such that $\langle\chi|\chi\rangle = 1$ has a place on the Bloch sphere.

$$\langle\mathbf{S}\rangle = \langle\chi|\mathbf{S}|\chi\rangle = \langle s|\mathbf{S}|s\rangle$$

Eigenmath script